

Knowledge Distillation for Edge-Friendly Plant Disease Classification

CENG 463 Final Project Progress Report

Group Members:

Abdulbaki Celtik (290201072)

Meric Kelemeden (290201091)

Department of Computer Engineering

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1 Problem Definition

The project studies whether knowledge distillation can produce a compact plant-disease classifier suitable for edge or mobile deployment. The target task is image-based classification on the PlantVillage dataset, where strong teacher models can achieve high accuracy but may be too large or slow for resource-constrained devices. The central research question is whether a MobileNetV3-Small student can approach the performance of larger ResNet teachers while reducing parameter count, model size, and inference latency.

The current report should be interpreted as a technical checkpoint. The implemented experiment uses the three-class PlantVillage Potato subset for fast validation of the data pipeline, model factory, evaluation code, and reporting workflow. These results are not claimed as final full-dataset performance.

2 Dataset and Preprocessing Status

The current experiment uses the PlantVillage Potato subset with three disease or health categories: Early Blight, Late Blight, and Healthy. The subset contains 1,721 RGB leaf images. Because the downloaded dataset does not provide official train, validation, and test splits, the pipeline creates a deterministic random split with seed 42.

Table 1: Dataset split sizes for the Potato subset.

Split	Samples	Fraction
Train	1,205	70%
Validation	258	15%
Test	258	15%

All images are resized to 224×224 pixels and normalized with ImageNet channel statistics.

Training uses random horizontal flipping and small random rotations, while validation and test transforms use only resizing and normalization. Class labels are remapped after filtering so that the loss function receives contiguous labels from 0 to $C - 1$.

3 Literature Review Progress

The literature review currently covers three project-relevant areas. First, Mohanty et al. [1] showed that convolutional neural networks can perform highly accurate image-based plant disease detection on PlantVillage, while also motivating later work on generalization beyond controlled image conditions. Second, Hinton et al. [2] introduced knowledge distillation, where a compact student model learns from both hard labels and softened teacher predictions. The planned distillation loss is

$$\mathcal{L} = \alpha T^2 \text{KL}\left(\sigma\left(\frac{z_s}{T}\right) \parallel \sigma\left(\frac{z_t}{T}\right)\right) + (1 - \alpha) \text{CE}(z_s, y), \quad (1)$$

where z_s and z_t are student and teacher logits, T is temperature, α is the distillation weight, and y is the ground-truth label. Third, MobileNet and MobileNetV3 [3, 4] provide efficient architecture choices for edge deployment because they reduce computation through depthwise separable convolutions and mobile-oriented building blocks.

4 Baseline Models

The codebase supports pretrained ImageNet backbones through the `torchvision` weights API. The final classifier layer is replaced to match the number of PlantVillage classes. The current status of the baseline and planned models is summarized in Table 2.

Table 2: Baseline and distillation experiment status.

Model	Role	Status
ResNet-18	Initial supervised baseline	Implemented
MobileNetV3-Small	Compact supervised baseline	In progress
ResNet-50	Strong teacher candidate	Planned
MobileNetV3 + KD	Distilled student	Next phase

The implemented ResNet-18 checkpoint uses cross-entropy loss, Adam optimizer, learning rate 10^{-3} , weight decay 10^{-4} , batch size 32, and 3 epochs. The configuration-driven training code automatically selects CUDA, Apple MPS, or CPU depending on the available hardware.

5 Initial Experimental Results

The current ResNet-18 experiment was read from the saved `resnet18` Potato-subset output directory. It uses a pretrained ResNet-18 with 11,178,051 trainable parameters and evaluates on the 258-image Potato test split. The saved metrics report 98.06% test accuracy, 95.90% macro-F1, 98.00% weighted-F1, and 1.72 ms single-image inference latency

on the recorded device. These results are strong for a three-class subset, but they should be treated as an initial pipeline checkpoint rather than evidence of final full-dataset behavior.

Table 3: Training history for ResNet-18 on the Potato subset.

Epoch	Train Loss	Val Loss	Val Acc.	Val Macro-F1
1	0.1734	0.0450	0.9806	0.9524
2	0.0768	0.0055	1.0000	1.0000
3	0.0546	0.0066	0.9961	0.9973

Table 4: Per-class classification report on the test set.

Class	Precision	Recall	F1-score	Support
Potato Early Blight	0.993	1.000	0.996	133
Potato Late Blight	0.962	0.990	0.976	102
Potato Healthy	1.000	0.826	0.905	23
Macro avg	0.985	0.939	0.959	258
Weighted avg	0.981	0.981	0.980	258

Table 5: ResNet-18 checkpoint summary from saved experiment outputs.

Metric	Value
Test accuracy	98.06%
Test macro-F1	95.90%
Test weighted-F1	98.00%
Trainable parameters	11,178,051
Inference latency	1.72 ms, batch=1
Epochs	3

6 Planned Improvements and Technical Direction

The next technical phase is to move from checkpoint validation to model comparison. First, MobileNetV3-Small will be trained as a supervised compact baseline. Second, ResNet-50 will be trained as a stronger teacher model. Third, MobileNetV3-Small will be trained with the distillation objective in Equation 1. Initial distillation settings will use $T = 4.0$ and $\alpha = 0.5$, then be tuned using validation macro-F1 and latency.

The comparison will include accuracy, macro-F1, weighted-F1, parameter count, model size, and batch-1 inference latency. If time permits, post-training INT8 quantization will be evaluated as an additional edge optimization method.

7 Ablation and Error Analysis Plan

The ablation plan is designed to isolate the contribution of architecture, distillation, and compression choices:

- Compare ResNet-18, ResNet-50, MobileNetV3-Small, and distilled MobileNetV3-Small under the same split and preprocessing pipeline.
- Vary distillation temperature T and soft-target weight α to measure sensitivity of the student model.
- Compare supervised MobileNetV3-Small against distilled MobileNetV3-Small to quantify the effect of teacher supervision.
- Measure the effect of optional INT8 quantization on accuracy and latency.

The error analysis will inspect misclassified test images, per-class precision and recall, confusion matrices, and confidence scores. Special attention will be given to the Potato Healthy class, because its small support makes recall more sensitive to a few errors.

8 Visualization and Interpretation

The current report includes the confusion matrix produced by the ResNet-18 checkpoint. This visualization helps identify whether errors are concentrated between visually similar disease categories or in the underrepresented healthy class.

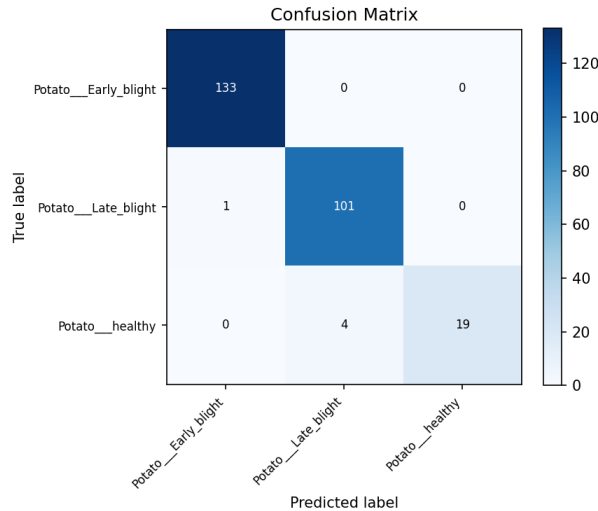


Figure 1: Confusion matrix for the ResNet-18 Potato-subset checkpoint.

Planned visualizations include training and validation curves, per-class metric bar charts, latency-versus-accuracy plots, and side-by-side examples of correct and incorrect predictions. These will be used to interpret whether a smaller student model fails uniformly or mainly on specific disease classes.

9 GitHub and Reproducibility Status

The project repository is available as a [public GitHub repository](https://github.com/bakiceltik/knowledge-distillation-edge). The implementation is organized around YAML configuration files, reproducible random seeds, and separate modules for data loading, model construction, training, evaluation, and distillation.

Repository URL:

```
https://github.com/bakiceltik/knowledge-distillation-edge
```

The current baseline can be reproduced with:

```
python -m src.data --config configs/baseline_resnet18.yaml
python -m src.train_baseline --config configs/baseline_resnet18.yaml
```

The saved checkpoint artifacts used in this report are the metrics JSON, results CSV, training-history CSV, model-summary JSON, and confusion-matrix PNG files from the ResNet-18 Potato-subset output directory.

10 Current Challenges and Next Steps

The main current challenge is that the Potato subset is small and visually controlled, so it may overestimate performance compared with the full 38-class PlantVillage task and real-world field images. The healthy Potato class is also much smaller than the disease classes, which makes class-level recall less stable. A second challenge is to ensure fair latency measurement across hardware backends, since CPU, MPS, and CUDA timings are not directly comparable unless evaluated under a fixed protocol.

The immediate next steps are to finish the MobileNetV3-Small supervised baseline, train the ResNet-50 teacher, run the first MobileNetV3 distillation experiment, and expand evaluation from the Potato subset toward the full PlantVillage dataset. The final comparison will focus on the accuracy, macro-F1, parameter, model-size, and latency trade-off between teacher, baseline, and distilled student models.

References

- [1] S. P. Mohanty, D. P. Hughes, and M. Salathe, “Using Deep Learning for Image-Based Plant Disease Detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [2] G. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” *NIPS Deep Learning Workshop*, 2015.
- [3] A. G. Howard et al., “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications,” *arXiv:1704.04861*, 2017.
- [4] A. Howard et al., “Searching for MobileNetV3,” *IEEE/CVF ICCV*, pp. 1314–1324, 2019.